PROJECT REQUIREMENTS

Password Strength Checker — Cybersecurity Mini Project

1. Hardware Requirements

Component	Minimum requirement
Computer	Desktop or laptop
Processor	Any processor capable of running Python 3
RAM	2 GB or more recommended
Storage	At least 100 MB of free space
Keyboard and display	Required for entering test passwords and viewing results

2. Software Requirements

Software	Requirement / purpose
Operating system	Windows or Linux; macOS can also be used
Programming language	Python 3
Code editor / IDE	IDLE, Visual Studio Code, or PyCharm
Python modules	re and getpass (both included with Python)
Execution environment	A terminal or command prompt that supports getpass

3. Functional Requirements

• Accept a password as user input.

• Check whether the password contains at least 8 characters.

• Check for uppercase letters, lowercase letters, numbers, and special characters.

• Calculate a score from 0 to 5 based on the five checks.

• Classify the result as Weak (0–2), Medium (3–4), or Strong (5).

• Display suggestions for any checks that are missing.

4. Non-Functional Requirements

• Usability: output should be simple and easy to understand.

• Performance: checks should finish almost immediately for normal password lengths.

• Privacy: the program should not save or transmit the password.

• Portability: should run on systems with Python 3 installed.

5. Installation and Execution

1. Install Python 3 from the official Python website if it is not already installed.

2. Save the source code in a file named password_checker.py.

3. Open a terminal or command prompt in the folder containing the file.

4. Run: python password_checker.py (some systems may use python3 password_checker.py).

5. Enter a made-up test password and review the displayed score and suggestions.

6. Security Notes

This is an educational project that checks basic password characteristics. It does not prove that a password is secure. Do not use real account passwords for demonstrations, and do not add password logging or plain-text storage. Length, uniqueness, and resistance to guessing are also important.